



NeXXtTimber[®]
by Timberlink

IT'S WHAT BETTER TOMORROWS ARE BUILT ON

Fire Performance Guide

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A modern interior space featuring a light-colored wooden ceiling and walls. A large black-framed glass door on the right provides a view of a green lawn, a wooden fence, and trees. A yellow armchair is positioned near the door. On the left, a white wall has a small framed picture of a tree. A red geometric overlay is in the bottom-left corner.

Fire Performance Guide

Important Information

The information within this document is based upon independent fire testing undertaken by Warringtonfire on NeXTimber® products, in accordance with AS 1530.4.

These tests, along with their assessment reports, conform with the requirements of the National Construction Code (NCC), Fire Resistance of Building Elements (NCC 2019 Volume One: Schedule 5 or NCC 2022 Volume One: Specification 7) and New Zealand Building Code (NZBC) - Acceptable Solution Clause C.

This guide must be read in conjunction with the NeXTimber® Technical Guide to obtain material properties and other relevant design information. This guide, technical information and test reports are specific to NeXTimber® products.

The formulation of NeXTimber® products (timber species, thickness and adhesive type) may differ from other manufacturers and the testing and performance results in this guide may differ from other manufacturer's products.

Application of this Fire Performance Guide by Suitable Professionals

This guide is intended for use by suitably qualified professionals to support the design and specification of NeXTimber® Cross Laminated Timber (CLT) and Glue Laminated Timber (GLT) products. Information and advice in this guide is of general nature only and has not been prepared with your specific project needs in mind. It is the responsibility of the user to ensure that the use of this guide and NeXTimber® products is appropriate to the relevant application and exercise their own professional judgement in doing so.

Suitably qualified professional shall comply with the intent of the relevant Australia Standards and all other relevant codes and industry best practice guidelines when assessing the use of NeXTimber® products. Overall building design issues are not considered in this guide. Users should always obtain specialist advice to ensure that the products referred to within this guide meet your specific requirements. For example, the structural engineer is responsible for undertaking the structural assessment of the overall building system to determine that load paths are maintained and that the structure can sustain the required fire exposure period, design loads and redistributed

load paths as part of an overall structural assessment.

NeXTimber® will not accept any liability for the failure of any other elements of a building which cause subsequent failure of a NeXTimber® product. Full responsibility for the design and compliance with the NCC and all relevant standards, codes and guidelines rests with the design professional specifying and certifying the product.

Penetrations and Interaction with Other Building Materials

The testing of NeXTimber® products has been carried out in accordance with AS 1530.4 and evaluates the fire-resistance performance of building components under specified conditions.

Additional testing may be needed to ensure compatibility and performance of penetrations, penetration treatments and interactions with other building materials (such as steelwork and concrete) in specific applications.

Users should consult with providers of fire protection systems for tested systems in CLT and consult with qualified professionals to verify compliance with all relevant standards and requirements for their particular projects.

NeXTimber® Technical Information Currency

This guide is current as at the date it bears. It is the user's responsibility to ensure that the latest version of this guide is in use at all times. Please check the NeXTimber® website for the latest version of this guide and other NeXTimber® documentation, including specific guide and test information.

This document shall be read in conjunction with Warringtonfire RIR report and where conflicting information is present, clarity shall be sought by the reader.

Access the latest technical information at www.nextimber.com.au

Acknowledgement of Contribution

NeXTimber® would like to acknowledge the following businesses who contributed to the contents of this document:

- Knauf Gypsum Pty Ltd
- TDA - Timber Development Association
- Warringtonfire

NeXTimber® Product Overview and Adhesive Performance

NeXTimber® CLT and GLT

NeXTimber® CLT and GLT are made from responsibly sourced Australian plantation pine.

Layers are bonded together using Melamine Formaldehyde (MF) adhesive (commonly referred to as Melamine) cured under pressure using an energy source.

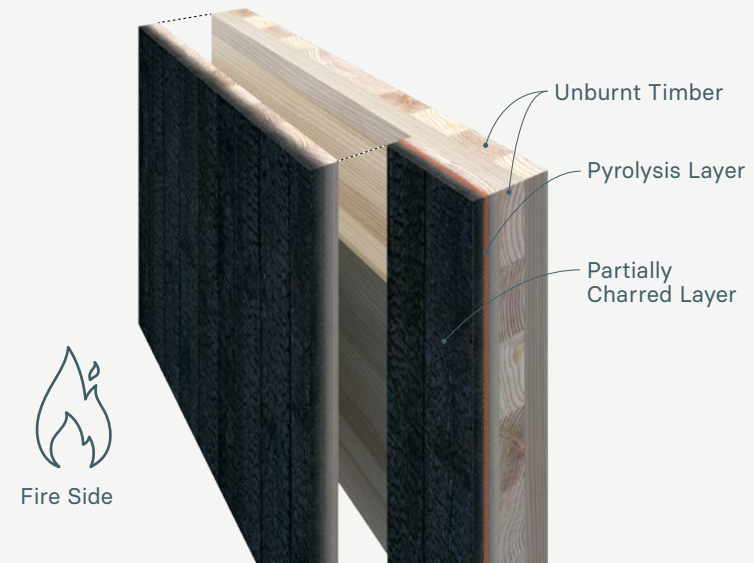
Melamine adhesives are common structural adhesives used in the manufacturing of timber products. They are thermoset polymers that harden during the curing process. Once hardened, this process is irreversible, resulting in an adhesive with high hydrolysis resistance (resistant to changing shape with exposure to heat), making it a well-suited adhesive for wood products¹.

NeXTimber® uses melamine for all joints and laminations, including finger joints, edge and face laminations. Melamine adhesives offer superior fire resistance compared to traditional polyurethane adhesives used in CLT manufacture, as melamine adhesives maintain thermal stability past the combustion point of the timber fibres, meaning that laminated products manufactured using melamine adhesives maintain their glue line integrity and will not delaminate as result of exposure to fire².

Warringtonfire test results demonstrate NeXTimber® products manufactured using high-performing, non-degrading, heat-resistant structural adhesives maintain glue line integrity and achieve a constant linear char rate; resulting in a predictable and reliable product with regard to fire performance, with comparable fire-resistant properties as solid timber products.

This linear charring behaviour allows the structural capacity to be confidently determined by analysing the residual section without the need of considering complex step-wise char rates associated with conventional polyurethane type adhesives due to delamination and char fall off.

Figure 1: Fire Exposed CLT Cross Section



¹Dagenais, C., Ranger, L., Benichou, N. and Su, J., 2021. Improved fire performance of cross-laminated timber. In World Conference on Timber Engineering (WCTE)-11th World Conference on Timber Engineering. Santiago, Chile (p. 10).

²Ebnesajjad, S. (2015) "Classification of Adhesives and Compounds," in [Adhesives Technology Handbook](#) ; Third edition

NeXTimber® Fire Testing

Products Used in Assessment

As there is a large range of fire resistance requirements for the various applications and uses for NeXTimber® CLT and GLT products, as well as the time it takes to conduct and report test and assess outcomes, additional Regulatory Industry Reports (RIR) will become available over time.

Refer to NeXTimber.com.au for updated RIR or contact our sales team directly.

Table 1. Summary of Products Used in Assessment

Product	Description
NeXTimber® CLT and GLT	Radiata Pine
Adhesive	Melamine Formaldehyde
Fire Rated Sealant	Bostik Fireban Acrylic
Fire-Grade Plasterboard	16mm Boral / Knauf Fire-Grade Plasterboard

Refer to fire reports for caulking details and fixing pattern of fire-protective plasterboard.

Figure 2: NeXTimber® CLT Wall Panel in Test Rig



Figure 3: NeXTimber® CLT Wall Panel at Test Completion



Figure 4: NeXTimber® CLT Floor Panel in Test Rig



Figure 5: NeXTimber® CLT Floor Panel at Test Completion



Fire Test Reports

The following tests, assessment and reports have been undertaken by Warringtonfire. The RIR is available upon request.

Table 2. Fire Reports

Report Number	Provider	Report Date	Revision
FRT210358	Warringtonfire	16/03/2022	R1.0
FRT210182	Warringtonfire	30/03/2022	R1.1
FRT210356	Warringtonfire	30/03/2022	R1.0
FRT210355	Warringtonfire	31/03/2022	R1.0
FRT210357	Warringtonfire	31/03/2022	R1.0
FRT210354	Warringtonfire	08/04/2022	R1.0
FRT210360	Warringtonfire	30/04/2022	R1.0

Table 3. Fire Assessments

Assessment Number	Provider	Report Date	Revision
FAS210228	Jensen Hughes	22/12/2025	R1.2

Table 4. Regulatory Information Report (RIR)

RIR Number	Provider	Report Date	Revision
FAS210228	Jensen Hughes	22/12/2025	RIR1.2

Fire Resistance Levels (FRL) or Fire Resistance Ratings (FRR)

The following is a general summary around the performance of Fire Resistance Levels (FRL), or Fire Resistance Ratings (FRR). Note that as per NZBC C/AS1 Appendix E, FRL is defined as Fire Resistance Rating (FRR) and can be used interchangeably with FRL herein. It is not advice and designers must seek their own professional advice and guidance.

Applying a Fire Resistance Level to Building Element

For NeXTimber® CLT and GLT, the fire resistance capacity has been determined from a series of fire tests conducted in different configurations with and without fire protective linings as set out in the 'Structural Performance and Testing' section.

All tests have been carried out in accordance with AS 1530.4 - Methods for Fire Tests on Building Materials, Components and Structures to determine the time to failure of the three performance measurements which combine to give an overall FRL (or FRR).

The FRL of bare and encapsulated CLT and GLT has been determined by tests and capacities calculated based on the performance of a prototype in the standard fire test and confirmed in a report prepared by Warringtonfire.

Based on our series of fire tests, Warringtonfire have undertaken an assessment and provided Warringtonfire's RIR, setting out the fire performance of NeXTimber® products.

Fire Resistance Level (FRL) or Fire Resistance Ratings (FRR)

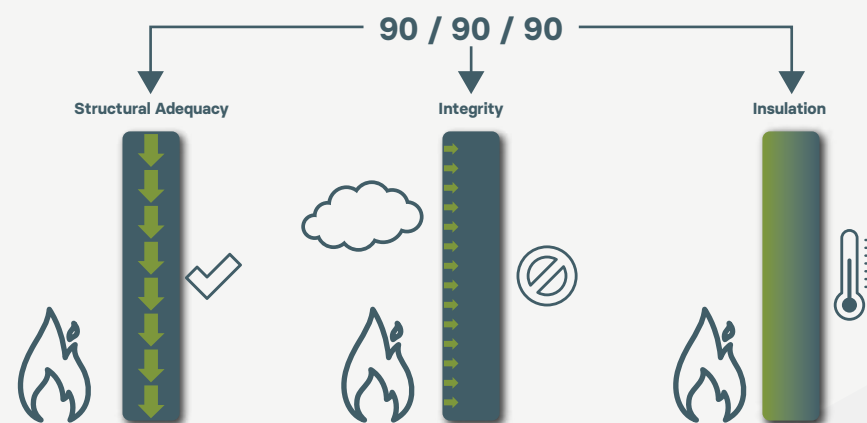
The FRL (or FRR) is determined by the system's ability to retain structural stability, avoid fire and smoke spread, provide safe conditions for evacuation, prevent concrete external walls from collapsing outwards in a fire, protect service and emergency equipment, and protect openings, penetrations, and fire brigade access.

For example, an FRL (or FRR) nominated as 90/90/90 indicates that the structure must sustain its performance for 90 minutes in the following three criteria as outlined in AS 1530.4:

- **Structural Adequacy:** Maintain strength and serviceability at the conclusion of the FRL period.
- **Integrity:** Prevent the passage of flames, smoke or hot gasses through the surface or joints.
- **Insulation:** Maintain an average increase in surface temperature on the non-fire side of less than 140°C and a peak increase of 180°C.

The FRL's (or FRR's) outlined within this document and Warringtonfire's Assessment and RIR are expressed in the format indicated in Figure 6.

Figure 6: Fire Resistance Level (FRL) or Fire Resistance Ratings (FRR)
FRL/FRR



Fire Resistance Levels (FRL) or Fire Resistance Ratings (FRR) Continued

Non-Combustible Building Elements Requirement

In certain applications, building codes often have a fire resistance requirement that requires materials to be non-combustible. Being a natural timber product, NeXTimber® CLT and GLT cannot be modified to meet this requirement.

Properties for Wall and Ceiling Linings

Limitations on material performance are expressed as group numbers. These performance ratings relate to the time it takes for a sample under tested conditions to reach flashover (transition from growth stage to full development across the surface of the fire compartment).

NeXTimber® CLT and GLT are manufactured entirely from Australian grown Radiata Pine, which achieves a Group 3 rating and with an average extinction area of less than 250m²/kg¹.

Where it is determined that performance is required to be better than Group 3, NeXTimber® CLT and GLT will require additional surface treatments (i.e. fire-grade plasterboard, non-combustible linings, intumescent coatings etc.).

In Australia, the use of intumescent coatings is not recognized under DTS provisions and will require a performance solution developed by the fire engineer and to the satisfaction of the building certifier.

Table 5. Fire Hazard Properties – Radiata Pine¹

Spread of Flame Index (0-10)	8
Smoke-Developed Index (0-10)	3

Table 6. Properties of Floor, Wall and Linings – Radiata Pine¹

Critical Radiant Flux (kW/m ²)	Greater than 2.2 and less than 4.5
Smoke Development Rate (% mins)	< 750

¹ Warringtonfire Assessment of Timber Floor, Wall and Ceiling Linings, against Building Code of Australia NCC Volume 1 Specification C1.10 RIR 411176

Structural Performance and Testing

NeXTimber® has undertaken testing to provide solutions for both bare wall and floor elements, as well as encapsulated in fire-grade plasterboard (FRPB).

CLT Residual Sections

Residual sections have been analysed and are outlined in Warringtonfire's Regulatory Information Report (RIR). They indicate the maximum permissible loads for the specimens under various loads, spans and FRL exposure. This report can be provided upon request.

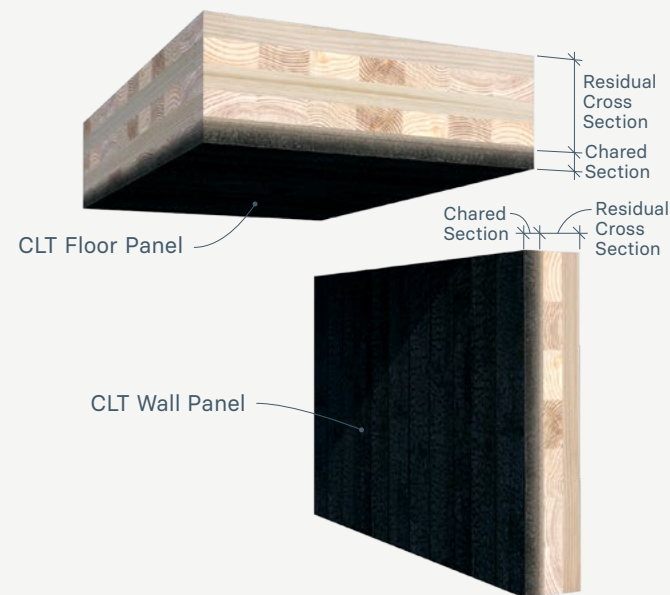
As part of this assessment, Warringtonfire have evaluated the stress within the panels for various FRL periods and load scenarios and provided maximum spans based on these loads.

The load applied equates to the maximum stress for the residual cross section in the test and subsequent assessment.

This stress is dependent on the effective length of the specimen, load eccentricities (accidental and load eccentricity to the centroid of the residual section) and the maximum load applied at the time of the relevant FRL.

Spans and loads outside those stated in the RIR will require further structural analysis to confirm that the stresses and deflections are not exceeded under the relevant loading conditions and combinations

Figure 7: Residual Cross Section



Structural Performance and Testing continued

Partially Protected/Encapsulated Timber

Testing demonstrates that wrapping or encapsulating the timber elements with passive fire protection such as FRPB provides an initial barrier, protects the timber and slows down the charring rate.

Depending on the level of encapsulation and duration of the fire, the underlying timber product may still experience some section loss and require a structural assessment of the remaining timber section.

NeXTimber® has undertaken testing of our products using single and double layers of 16mm FRPB and provides charring models within this guide for these systems.

Designers shall note that the screw pattern and screw length are essential to achieving the FRL performance of the product, refer to the relevant encapsulating drawings for standard fixing details as shown in Figures 18 & 21-24.

Note: Encapsulated indicates protected with 1 layer of 16mm Fire grade plasterboard, secured in accordance with the relevant details in this Guide.

Figure 8: Encapsulated NeXTimber® CLT Panel

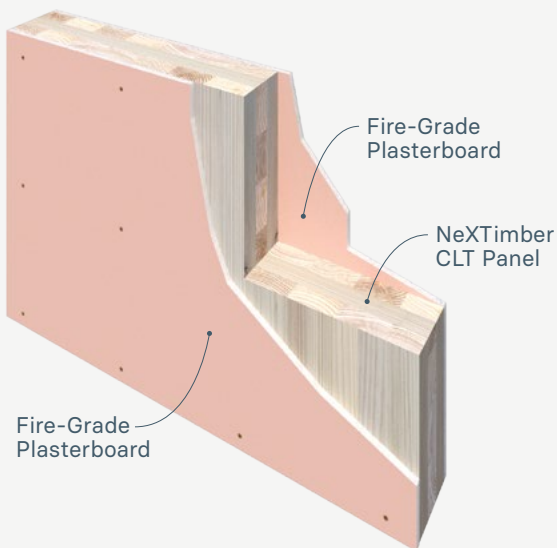
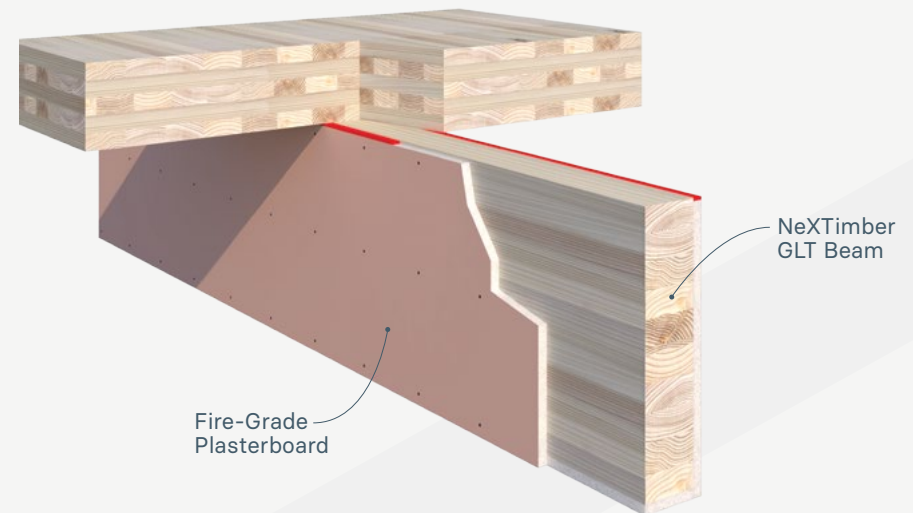


Figure 9: Encapsulated NeXTimber® GLT



Fire Performance – NeXTimber® Wall and Floor Panels

The below tables indicate char rates and char depths of NeXTimber® CLT as determined by test and assessment carried out by Warringtonfire and summarized in their RIR.

Table 7. NeXTimber® CLT Panels – Notional Char Rates Determined by Test

	FRP (mins)	Char Rate (mm/min)
Unprotected Floors and Walls	0–120	0.80

Table 8. NeXTimber® CLT Panels – Notional Char Rates Determined by Test

	Time Period (mins)	Char Rate (mm/min)
Floors and Walls protected with 1 layer 16mm Fire Grade Plasterboard	0 – 30	0.00
	30 – 120	0.35
	120 – 180	0.80

Figure 10: NeXTimber® Notional Char Rates – CLT

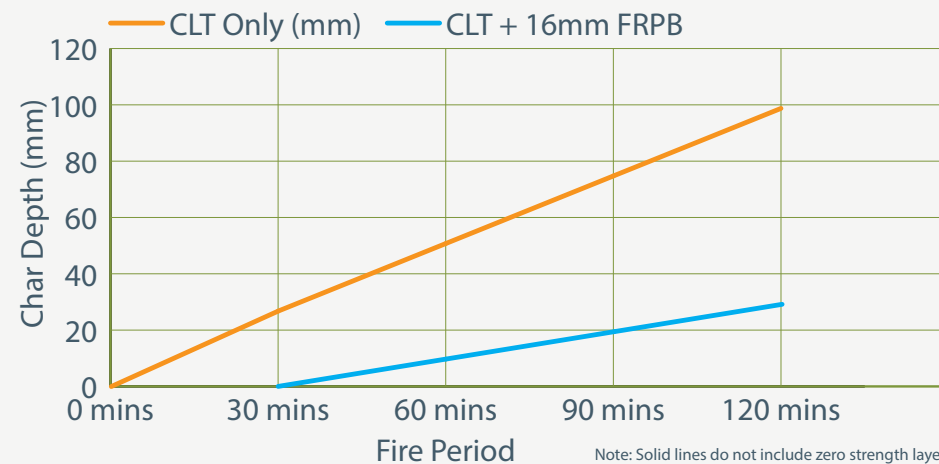


Table 9. Integrity and Insulation - Minimum Initial Panel Thickness (mm)

Time Period (mins)	Wall Joints		Floor Joints		
	Half Lap Unprotected	Half Lap Encapsulated	Half Lap Unprotected	35mm Spline Unprotected	Half Lap or 35mm Spline Encapsulated
30	90	90	90	90	90
60	90	90	90	110	90
90	180	90	180	130	90
120	230	100	230	150	100
180	-	200	-	-	200

Notes:

- The FRL is only valid if the professional or project structural engineer confirms that the NeXTimber CLT element satisfies the structural adequacy requirements.
- Spline plate connections not permitted on fire side of wall to wall connections.
- Encapsulated indicates protected with one layer of 16mm fire grade plasterboard secured in accordance with the requirements of the test report.

Calculated Char Depths – From Calculated Char Rates

The effective depth of charring (d_c) after a period of time (t) shall be calculated as follows;

$$d_c = \Sigma(c \times \Delta t) + 7mm$$

Where;

d_c = depth of char

c = notional charring rate for interval, in millimeters per minute (refer to Figure 10)

Δt = time interval, in minutes

The 7mm accounts for a layer of uncharred timber at elevated temperatures, which is assumed to contribute no strength or stiffness to the residual section.

To confirm structural adequacy, the residual section should be assessed by the structural engineer by considering the initial profile, then subtracting both the char depth and the layer of zero strength.

Char Depths Below Splines and Lap Joints

When using the calculation method based on notional char rates determined from test, the calculated char depth must maintain a minimum thickness of 20mm from the underside of the joint to the char layer.

$$t_{min} = \Sigma(c \times \Delta t) + 20mm$$

For further information, refer to 5.3.2 of Jensen Hughes fire Assessment Report

Figure 11: CLT Spline Joint

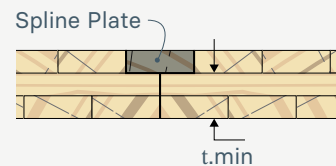
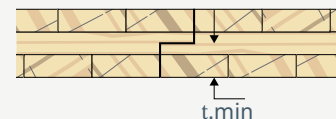


Figure 12: CLT Lap Joint



Fire Rated Panel Joints and Design Considerations

Panel to Panel Joints

Joint detailing may often dictate the FRL of the floor or wall system, and suitably qualified professional advisors must consider the fire performance of joints when prescribing the type of joints and quality of installation within the building elements. Panels shall be tightly fitted and screwed together with caulking placed inside the joint to maintain insulation and integrity and sufficient cover to the joint needs to be maintained, and joints and gaps in panels need to be filled to ensure integrity and insulation are also maintained.

Floor to floor joints can be assembled with both a lap joint and spline joint, however they cannot be used interchangeably under all circumstances due to their differing fire performance.

NeXTimber®'s fire assessment report dictates when a spline must be used over a lap joint in order to achieve the fire period nominated in Table 9 and the Regulatory Information Report (RIR) and corresponding fire assessment.

Typically, this requirement is confined to thinner panels as there needs to be adequate separation from the fire source to the underside of the joint. separation from the fire source to the underside of the joint. If a lap joint is preferred, it may be necessary for the designer to specify a thicker panel to achieve this requirement. This is generally only required for thin panels.

Spline plate connections are currently not permitted in NeXTimber® wall-to-wall connection details where a FRL is required.

Interfaces with Other Materials

Interfaces with other materials or gaps are generally protected with intumescent strips, mineral wool, fire rated caulking or board products.

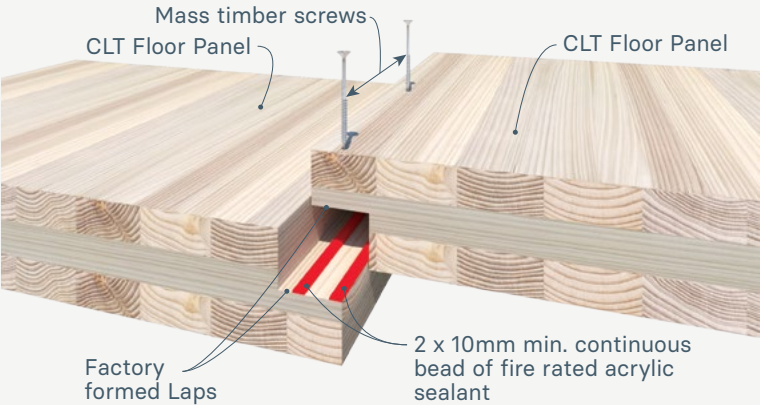
Additional fire stopping solutions may be provided following testing and certification with passive fire protection suppliers and made available through nextimber.com.au.

This guide does not contain all assembly details encountered within a building. NeXTimber® will be undertaking ongoing testing for various connection and interface details. This guide will be updated when these details are available.

Please contact our technical team should you require specific details and visit nextimber.com.au for regular updates.

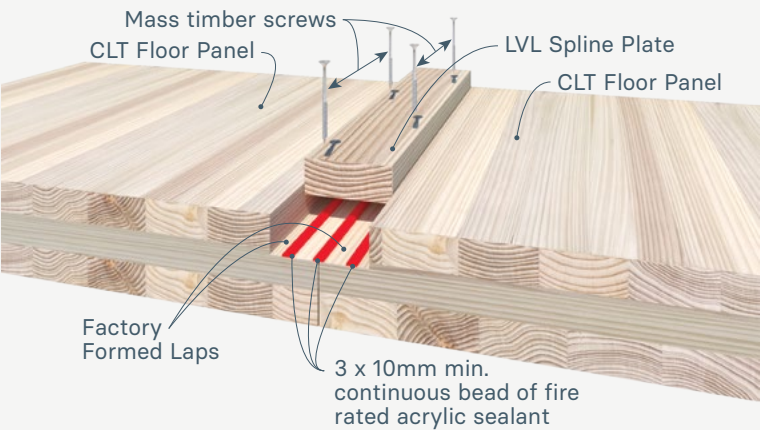
Fire Rated Panel Joints and Design Considerations Continued

Figure 13: Typical NeXTimber® CLT Floor Lap Joint



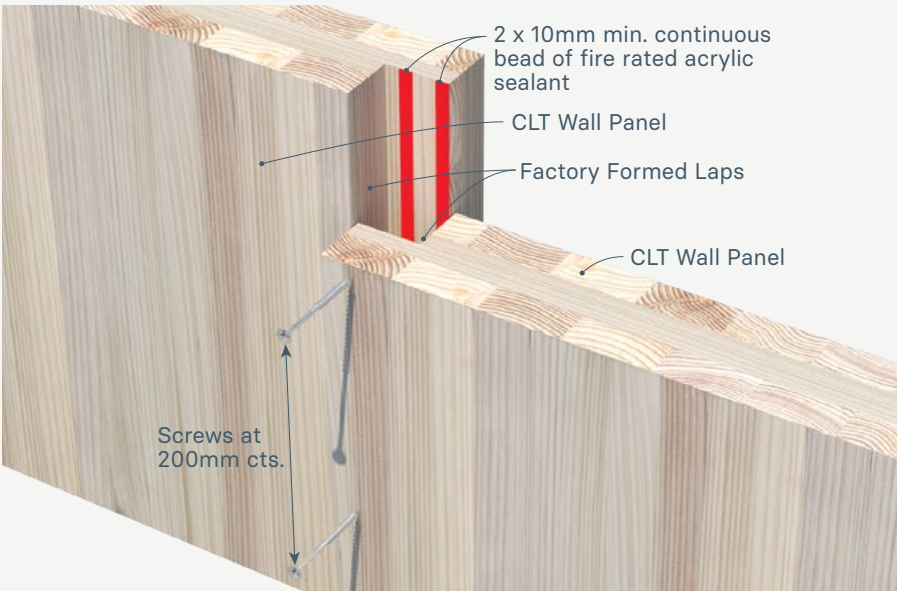
Note: Detail still applicable if using FRPB

Figure 14: Typical NeXTimber® CLT Floor Spline Joint



Note: Detail still applicable if using FRPB

Figure 15: Typical NeXTimber® CLT Wall Lap Joint



Note: Detail still applicable if using FRPB

Additional Design Considerations and Detailing

Below are some design considerations that engineers, architects and builders should consider as part of the structural detailing and assembly to achieve adequate fire performance. The below list is not exhaustive and professional advice should be sought when finalising the structural details.

Design Considerations

In addition to the assessment of the panel itself, designers will need to consider the overall performance of the building structure, particularly detailing connection performance. This includes but is not limited to:

- Maintaining sufficient fixity, specifically for horizontal spanning members supporting floors;
- Accounting for additional actions caused by the load path and reduction in cross-section of the element;
- Carrying out a systematic risk assessment for accidental loads, including fire, when using CLT in
- Multi-storey buildings as this can also help to identify and remove risks;
- Understanding that horizontal spanning elements, including deep beams, headers and lintels, will behave differently given the alternating grain orientation for horizontal spanning members and loss of fibre; and
- Maintaining joints to ensure there is a sufficient level of protection to prevent the onset of flame and smoke penetration.

Connections

Engineers and detailers need to exercise care around the detailing of connections and joints to account for material loss. With both exposed or partially protected timber products, metallic connections and fasteners may need additional protection. This includes but is not limited to:

- Encase fixings in timber by overdriving and providing solid timber glued plugging of equal or greater density than the primary element;
- Provide sufficient timber encapsulation to internal structural plates and fittings;
- Introduce additional passive fire protection (FGPB and other non-combustible insulating materials); and
- Provide additional connections and/or structural redundancy in the event that connections are compromised or designed for robustness.
- Refer to AS/NZS 1720.4 for additional guidance.

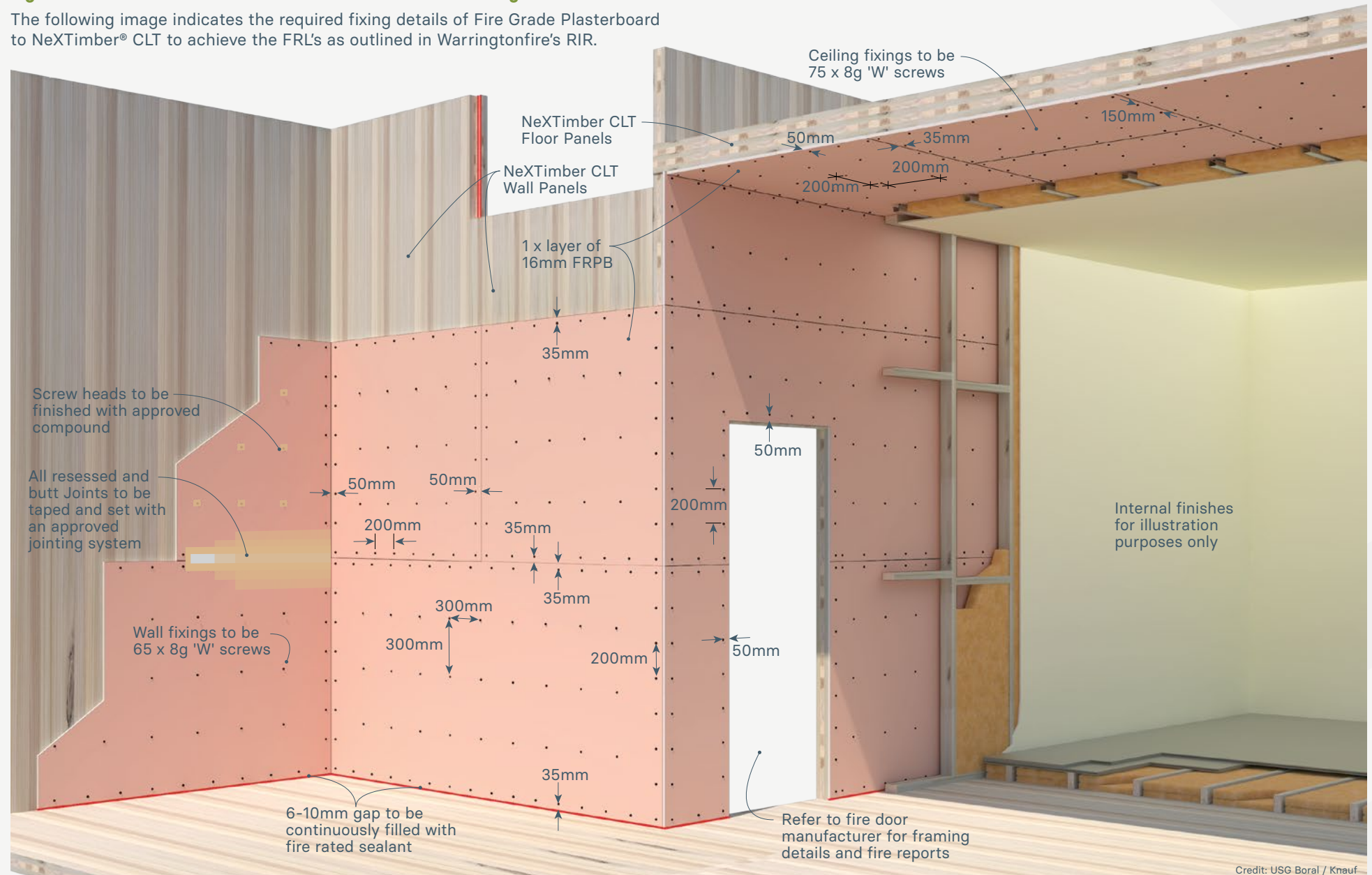
Floor and Wall Penetrations

Floor and wall penetrations require specific protection to maintain separation. There are numerous tested systems available in the market that have been tested with, and are compatible with CLT. NeXTimber® can partner with fire protection manufacturers to support testing and provide tested solutions. Please contact our technical team should you require specific details..



Figure 16: NeXTimber® CLT Panel – Fire Grade Plasterboard Fixing Details

The following image indicates the required fixing details of Fire Grade Plasterboard to NeXTimber® CLT to achieve the FRL's as outlined in Warringtonfire's RIR.



Credit: USG Boral / Knauf

Predicted NeXTimber® GLT Char Depths and Rates

The below table indicates char rates and char depths of NeXTimber® GLT as determined by test and assessment carried out by Warringtonfire and summarized in their RIR.

Table 10. Predicted Effective Char Depth - NeXTimber® GLT

Time	NeXTimber® GLT Only (mm)	NeXTimber® GLT + 1 Layer 16mm Fire grade Plasterboard (mm)	NeXTimber® CLT + 2 layers 16mm Fire Grade Plasterboard (mm)
30 mins	31.0	7.0	7.0
60 mins	55.0	22.0	7.0
90 mins	79.0	46.0	7.0
120 mins	103.0	70.0	31.0

Notes:
Char Depths Include 7mm 'Zero Strength Layer' as Defined in AS 1720.4.
Refer to Figures 17-20 for Fire Grade Plasterboard fixing details to NeXTimber® GLT

Residual Cross-Section

For NeXTimber® GLT, the structural capacity can be determined by calculating the residual cross-section (determined using the above char rates in Table 10) and the actions obtained from load combination outlined in AS/NZS 1170.0 and evaluating using the methods outlined in AS 1720.1.

NeXTimber® GLT Section Sizes

Refer to the NeXTimber® Technical Guide for NeXTimber® standard sizes and cross sections.

Refer to AS/NZS 1720.4 for limitations on use when using notional char rates determined by test.

Figure 17: Notional Char Rates – GLT

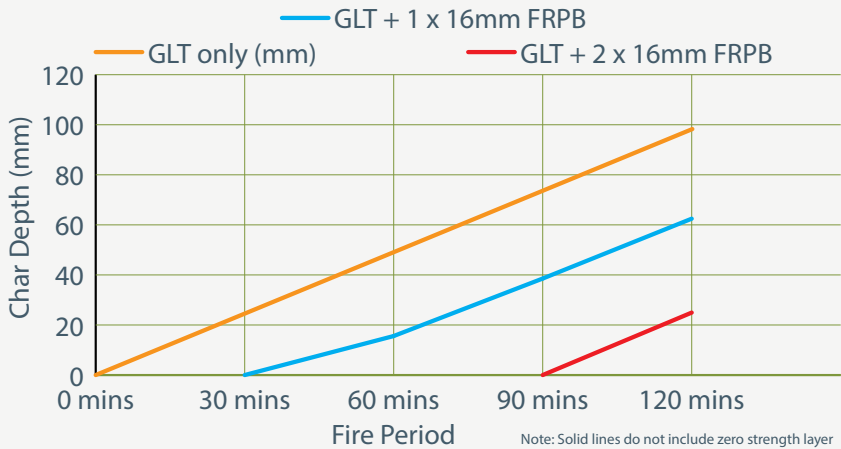
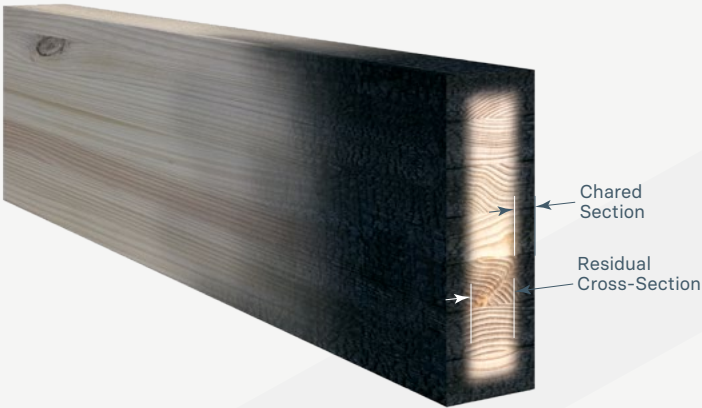


Figure 18: GLT Residual Cross-Section



NeXTimber® GLT Column – Fire Grade Plasterboard Fixing Details

The following images indicate the required fixing pattern and specification of Fire Grade Plasterboard to NeXTimber® GLT to achieve the FRL's as outlined in Warringtonfire RIR.

Figure 19: Typical NeXTimber® GLT Column Encapsulated with 1 Layer FRPB

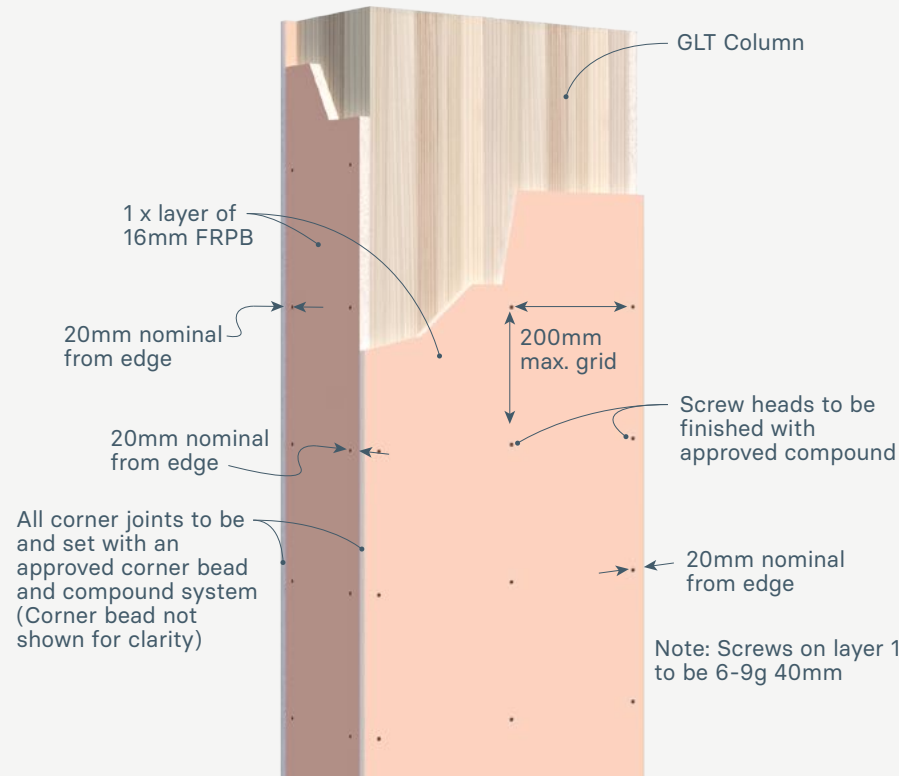
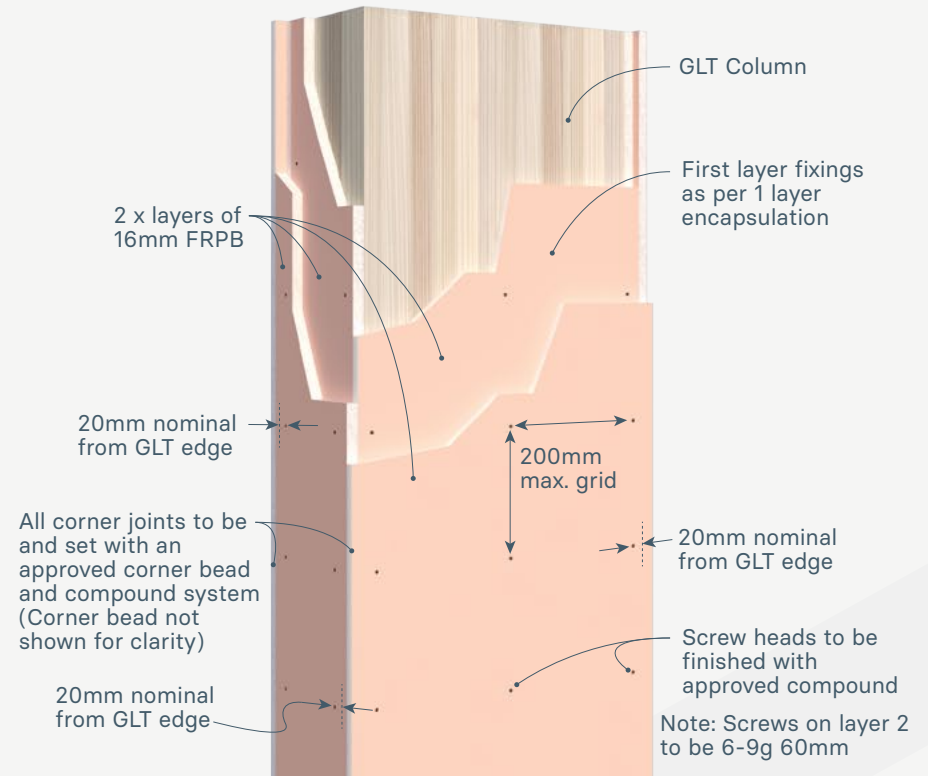


Figure 20: Typical NeXTimber® GLT Column Encapsulated with 2 Layer FRPB



NeXTimber® GLT Beam – Fire Grade Plasterboard Fixing Details

Figure 21: Typical GLT column Encapsulated with 1 Layer of FRPB

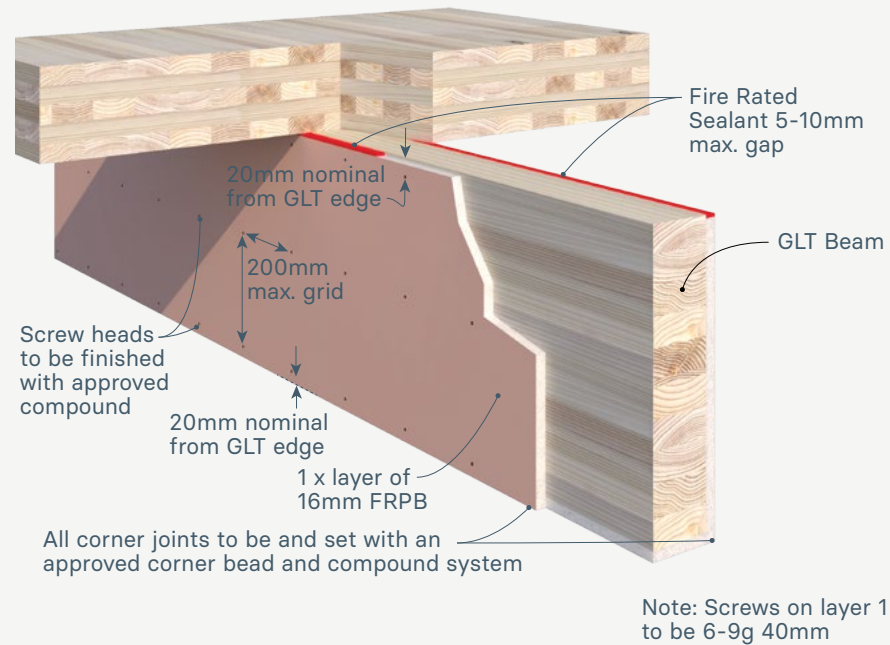
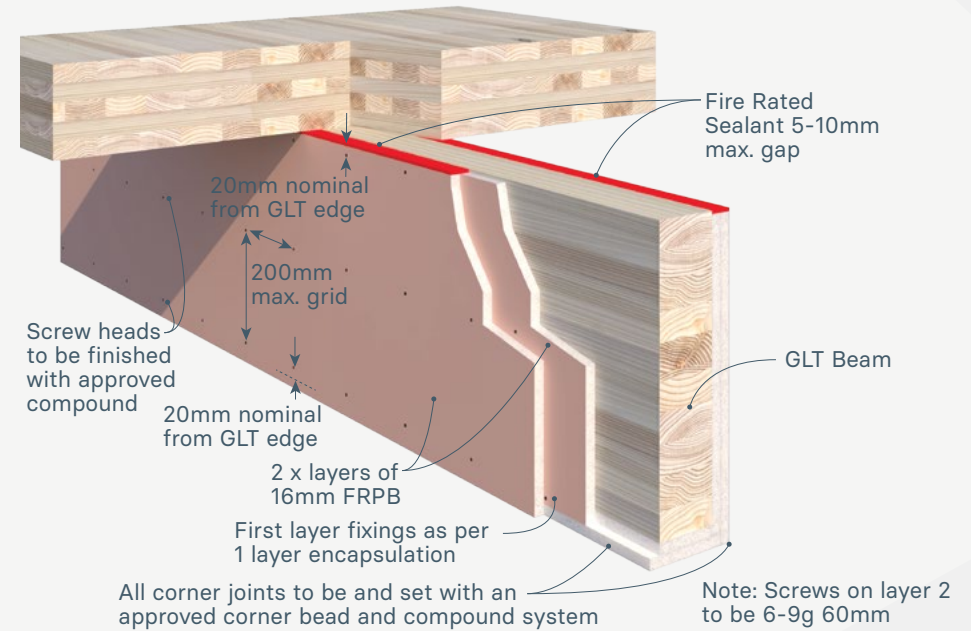


Figure 22: Typical GLT column Encapsulated with 2 Layers of FRPB



It starts with a seed from nature, with soil,
water and sunshine added.

It's what a better tomorrow will be built on,
built out of, built with.

It's what the craftsmen, the draftsmen,
the architect, the engineer – all those who
shape tomorrow – should lean on.

If it can be renewable and store carbon, it
should, and it can.

And if it can all be done quicker, quieter,
smarter, and safer it should.

And it can.

You see, if we treat our buildings as living
breathing things, then the living breathing
things that inhabit them, will have a future
to look forward to.

Everything we do today matters, for
tomorrow will be where we all reside.

However you use NeXTimber®, you won't
just be building a building.

You'll be creating a better future.

NeXTimber®
by Timberlink

IT'S WHAT BETTER TOMORROWS ARE BUILT ON

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Version 3.0
June 2026

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CLT and GLT illustrations
used throughout are
artist impressions

